Shuguang Chen

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PROFESSIONAL SUMMARY

Staff Applied Scientist with a PhD in Natural Language Processing. Specialize in large language models (LLMs), Agentic AI, function calling, and NLP-driven automation. Proven ability to bridge cutting-edge research with real-world enterprise applications, delivering scalable and efficient AI systems. Strong publication record at top NLP venues, complemented by industry experience deploying advanced AI frameworks and models into production.

EDUCATION

University of Houston — *Houston, TX*

Ph.D. in Computer Science — *Aug. 2018 - Dec. 2022*

- Advisor: Prof. Thamar Solorio
- Research Area: Natural Language Processing (NLP)
- Dissertation: Named Entity Recognition on Social Media

Beijing Forestry University — *Beijing, China*

B.S. in Computer Science — *Sep. 2014 - Jul. 2018*

- Advisor: Prof. Meng Wei
- Thesis: Music Generation Using Recurrent Neural Networks

WORK EXPERIENCE

Staff Applied Scientist — *Seattle, WA*

DigitalOcean Holdings, Inc. — *Mar. 2026 - Present*

Led the development of enterprise LLM orchestration and optimization frameworks for complex multi-step workflows, with a focus on reliable agent execution and post-deployment improvement.

- **Plano-Orchestrator (LLM Orchestration)**: Developed and advanced a family of orchestration models for multi-step agent workflows, designed to improve planning, tool coordination, and end-to-end task completion.
- **Signals for Agentic Interaction Triage**: Proposed a lightweight, signal-based framework for triaging agent trajectories using structured interaction and execution signals, showing 82% informativeness and a 1.52× efficiency gain over random sampling in controlled annotation studies.

Senior Applied Scientist — *Bellevue, WA*

Katanemo Lab, Inc. — *May. 2024 - Mar. 2026*

Led the development and release of a suite of open-source models for Agentic AI, accumulating over 200K downloads in the open-source community and deployed in enterprise applications.

- **Arch-Agent (Multi-turn Agentic AI)**: Engineered a family of models for complex, multi-turn workflows, achieving top-tier performance and ranking 3rd out of 116 models on the Berkeley Function-Calling Leaderboard. Designed for real-world agent deployments with advanced decision-making, error recovery, and contextual continuity.
- **Arch-Function (LLM Function Calling)**: Developed state-of-the-art function calling models with performance on par with GPT-4. Built a model suite supporting single, parallel, and multiple function calls with 95.12% hallucination resistance, optimized for low-latency, high-throughput production environments.
- **Arch-Router (Intelligent LLM Routing)**: Achieved a 20x cost reduction by developing a preference-based routing model that enables transparent and controllable model selection. Architected a flexible framework to power multi-LLM environments without router retraining.

Postdoctoral Researcher — *West Lafayette, IN*

Purdue University, PMML Lab — *Aug. 2023 - Apr. 2024*

- Conducted interdisciplinary research integrating NLP and mathematics.
- Developed specialized training methods to enhance LLM mathematical reasoning.
- Collaborated with mathematics faculty on AI-driven problem-solving.

SELECTED PROJECTS

Style Transfer as Data Augmentation in NLP [Github] — *Houston, TX*

Affiliation: University of Houston, Advisor: Prof. Thamar Solorio — *Nov. 2021 - Aug. 2022*

- Developed a novel approach to map text styles across domains.
- Enhanced domain adaptation by transferring stylistic attributes in NLP tasks.
- Improved data augmentation techniques for low-resource settings.

Multimodal Named Entity Recognition on Social Media [Github] *Houston, TX*
Advisor: Prof. Thamar Solorio *Sep. 2019 - Sep. 2021*

- Research Topic: Multimodality and Language Grounding to Vision
- Studied multimodal information representation, extraction, and fusion.
- Proposed methods to reduce performance degradation by fusing visual and textual information.

Machine Translation on Code-switched Data *Houston, TX*
Advisor: Prof. Thamar Solorio *Jun. 2020 - Dec. 2020*

- Conducted research on machine translation in linguistic code-switching.
- Created new standard datasets in different language pairs and provided strong baseline systems.

Named Entity Recognition on Diachronic Twitter Data [Github] *Houston, TX*
Advisor: Prof. Thamar Solorio *Jun. 2020 - Apr. 2021*

- Research Topic: Information Extraction
- Probed the impact of temporal drift on entity memorization and context generalization.
- Presented a method to efficiently update model parameters with the most informative data.

SELECTED PUBLICATIONS

Full list of publications available at Google Scholar.

- **Signals: Trajectory Sampling and Triage for Agentic Interactions**
  Shuguang Chen, Adil Hafeez, Salman Paracha. *arXiv:2604.00356*
- **Arch-Router: Aligning LLM Routing with Human Preferences**
  Co Tran, Salman Paracha, Adil Hafeez, **Shuguang Chen**. *arXiv:2506.16655*
- **LLM Reasoning Engine: Specialized Training for Enhanced Mathematical Reasoning**
  Shuguang Chen, Guang Lin. *NAACL 2025 (KnowledgeNLP)*
- **Context-aware adversarial attack on named entity recognition**
  Shuguang Chen, Leonardo Neves, Thamar Solorio. *EACL 2024 (W-NUT)*
- **Style Transfer as Data Augmentation: A Case Study on Named Entity Recognition**
  Shuguang Chen, Leonardo Neves, Thamar Solorio. *EMNLP 2022*
- **CALCS 2021 Shared Task: Machine Translation for Code-Switched Data**
  Shuguang Chen, Gustavo Aguilar, Anirudh Srinivasan, Mona Diab, Thamar Solorio. *NAACL 2021 (CALCS)*
- **Data Augmentation for Cross-Domain Named Entity Recognition**
  Shuguang Chen, Gustavo Aguilar, Leonardo Neves, Thamar Solorio. *EMNLP 2021*
- **Can images help recognize entities? A study of the role of images for Multimodal NER**
  Shuguang Chen, Gustavo Aguilar, Leonardo Neves, Thamar Solorio. *EMNLP 2021 (W-NUT)*

HONORS & AWARDS

2022	**Cullen Graduate Student Success Fellowship**, University of Houston	*Houston, TX*
2018	**Outstanding Graduate Awards**, Beijing Forestry University	*Beijing, China*
2018	**Academic Merit Scholarship**, Beijing Forestry University	*Beijing, China*
2016	**Bronze Medal**, China Collegiate Programming Contest (CCPC)	*Hangzhou, China*
2016	**2nd prize**, The 7th Blue Bridge Cup National Software Competition	*Beijing, China*

PROFESSIONAL SERVICE

- **Program Committee Member**, ACL, EMNLP, NAACL, W-NUT, and ARR.
- **Co-organizer**, The Fifth workshop on Computational Approaches to Linguistic Code-Switching [**CALCS**].
- **Webmaster**, Linguistic Code-switching Evaluation Benchmark [**LinCE**]

TECHNICAL SKILLS

- **Programming Languages:** Python, C/C++
- **ML/AI Frameworks:** PyTorch, Transformers, Hugging Face, DeepSpeed, Triton.
- **Areas of Expertise:** LLMs, NLP, function calling, agentic AI, adversarial robustness, data augmentation.
- **Tools & Systems:** Distributed training, model serving, performance optimization, MLOps Pipelines